

Calculus II

Integrals of the form $\int \sin^n x \cos^m x dx$, theory

Todor Milev

2019

$$\int \sin^m x \cos^n x dx$$

When n – odd:

$$\int \sin^m x \cos^n x dx$$

When m – odd:

$$\int \sin^m x \cos^n x dx = \int \sin^m x \cos^{n-1} x d(\sin x)$$

When n – odd:
 $\cos x dx$
 $= d(\sin x)$

$$\int \sin^m x \cos^n x dx$$

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$$\begin{aligned}
 \int \sin^m x \cos^n x dx &= \int \sin^m x \cos^{n-1} x d(\sin x) \\
 &= \int \sin^m x (1 - \sin^2 x)^{\frac{n-1}{2}} d(\sin x)
 \end{aligned}$$

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If both m, n – even, use $\left| \begin{array}{l} \sin^2 x = \frac{1 - \cos(2x)}{2} \\ \cos^2 x = \frac{\cos(2x) + 1}{2} \end{array} \right.$ and substitute $s = 2x$ to lower trig powers. Repeat above considerations.

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